

System Call Investigation

A Beginner-Friendly Quiz with AI

Course

Operating Systems

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What you will do

You will read a short C program and two simplified Linux traces. Trace A shows a successful run. Trace B shows a failed run. You will explain what happened, ask AI for help, check the AI response against the evidence, and submit a short report.

What you need

- This quiz document.
- A web browser and an approved AI assistant.
- Word or Google Docs for your report.

Important rules

- You do not need to install or run any program.
- AI may help you understand the evidence, but your final answers must be in your own words.
- Every answer must mention the exact program line or trace line that supports it.
- Include the AI conversation at the end of your report.

Evidence 1 The C program

The program opens test.txt, reads its contents, displays those contents on the screen, and closes the file. Focus on Lines 7, 11, 16, and 25.

```
1 #include <stdio.h>
2 #include <fcntl.h>
3 #include <unistd.h>
4 #include <errno.h>
5
6 int main(void) {
7     int fd = open("test.txt", O_RDONLY);
8     if (fd == -1) { perror("open"); return 1; }
9
10    char buffer[100];
11    ssize_t n = read(fd, buffer, sizeof(buffer));
12    if (n == -1) { perror("read"); close(fd); return 1; }
13
14    ssize_t written = 0;
15    while (written < n) {
16        ssize_t result = write(STDOUT_FILENO,
17                               buffer + written, n - written);
18        if (result == -1) {
19            if (errno == EINTR) continue;
20            perror("write"); close(fd); return 1;
21        }
22        written += result;
23    }
24
25    if (close(fd) == -1) { perror("close"); return 1; }
26    return 0;
27 }
```

Words you need to know

Term	Simple meaning
System call	A request from a program to the operating system kernel.
File descriptor	A small number the operating system uses to identify an open file or stream.
Descriptor 1	Standard output: normally the screen.
Descriptor 2	Standard error: normally error messages on the screen.
Return value	The result shown after the = sign in a trace.
ENOENT	The requested file or path does not exist.

Evidence 2 Trace A Successful run

For this run, test.txt exists. It contains: Hello, OS Kernel! followed by a newline.

```
A01 execve("./read_file", ["/read_file"], ...) = 0
A02 mmap(NULL, 53198, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f2a1c000000
A03 openat(AT_FDCWD, "test.txt", O_RDONLY) = 3
A04 read(3, "Hello, OS Kernel!\n", 100) = 18
A05 write(1, "Hello, OS Kernel!\n", 18) = 18
A06 close(3) = 0
A07 exit_group(0) = ?
A08 +++ exited with 0 +++
```

How to read Trace A

Line	What it shows
A03	test.txt is opened. The operating system returns descriptor 3.
A04	The program reads 18 bytes from descriptor 3.
A05	The program writes those 18 bytes to descriptor 1, the screen.
A06	The program closes descriptor 3 successfully.

Evidence 3 Trace B Failed run

For this run, test.txt does not exist.

```
B01 openat(AT_FDCWD, "test.txt", O_RDONLY) = -1 ENOENT
B02 write(2, "open: No such file or directory\n", 32) = 32
B03 exit_group(1) = ?
B04 +++ exited with 1 +++
```

What Trace B shows

B01 fails because test.txt is missing. The program therefore prints an error message to descriptor 2 and stops. It cannot read or close test.txt because the file was never opened successfully.

Required AI activity

Copy Trace A and Trace B into an approved AI assistant. Paste the prompt below. Save the response for your report.

Explain Trace A and Trace B in simple language. For Trace A, explain Lines A03–A06 and the return value on each line. For Trace B, explain Lines B01–B04 and the meaning of ENOENT. Explain file descriptors 1, 2, and 3. Finally, explain the difference between entering kernel mode for a system call and switching the CPU to another process. Do not write my final quiz answers.

Check the AI response

- Does it say A03 returns descriptor 3?
- Does it say A04 reads 18 bytes?
- Does it say A05 writes to descriptor 1, the screen?
- Does it say B01 fails because test.txt is missing?
- Does it avoid claiming that every system call switches the CPU to another process?

Question 1: Successful execution (25 points)

Using A03–A06, explain what happens to test.txt. Write one sentence for each line:

- A03: What is opened, and what descriptor is returned?
- A04: Which descriptor is read, and how many bytes are read?
- A05: Where are the 18 bytes written?
- A06: Which descriptor is closed, and did close() succeed?

Question 2: Failed execution (20 points)

Using B01–B04, answer all four parts:

- Why does openat() fail in B01?
- What does -1 ENOENT mean?
- Why is the error message written to descriptor 2 in B02?
- Why are there no successful read() or close() calls for test.txt after B01?

Question 3 Program and trace connection (20 points)

Match each program line to the trace line that shows the same operation. Then briefly explain the match.

Program line	Operation	Trace line to identify
Line 7	Open test.txt	_____
Line 11	Read from test.txt	_____
Line 16	Write the file contents	_____
Line 25	Close test.txt	_____

Question 4 Kernel mode and context switch (20 points)

Answer each part in 1–2 sentences:

- Which operations in Trace A require help from the kernel? Give at least two line numbers.
- Why must a program enter kernel mode to open or read a file?
- Why does entering kernel mode not necessarily mean that the CPU starts running a different process?
- What can strace show, and what can it not directly show about the CPU mode bit or a context switch?

Question 5 Evaluate the AI response (15 points)

Choose one specific statement from the AI response. Complete all three steps:

- Copy or summarize the AI statement.
- Say whether you verified it, corrected it, or made it more precise.
- Give the exact trace line that supports your decision.

Submission instructions

- Write approximately 1–2 pages, not counting the AI transcript.
- Use the five question titles as headings.
- Add your name, Panther ID, course section, and date.
- Append a screenshot or transcript of the AI conversation.
- Save the file as Lastname_Firstname_SystemCallQuiz.pdf.
- Upload the single PDF to iCollege before time expires.

Total 100 points

The points for each question are shown directly in its title.